

Astro 331, Lab 5

1) Active Array Size = 2048 px ~~1536 px~~

Focal Length = 2.8 mm

Aperture Diameter = 1.0 mm

FOV = 75° (H) · 56.4° (V)

Sensor Size = 3.6 mm (H) · 2.7 mm (V)

only using horizontal

$$Res = \frac{2.44 \lambda h}{D}$$

Google search yielded ~500 nm

for average wavelength of visible

light, so assume $\lambda = 5 \cdot 10^{-7} \text{ m}$

$$= \frac{2.44(5 \cdot 10^{-7}) \cdot h}{(1 \cdot 10^{-3})}$$

$$Res_{\text{diffraction}} = 1.22 \cdot 10^{-3} h$$

$$mag = \frac{f}{h} = \frac{f}{R} \Rightarrow R = \frac{ch}{f}$$

need 2R

$$s_{\text{max}} = 2R = \frac{2ch}{f} = \frac{3.6h}{2.8} = 1.286h$$

$$Res_{\text{pixel}} = \frac{2R}{\# \text{ pixels}} = \frac{1.286h}{2048}$$

$$Res_{\text{pixel}} = 6.279 \cdot 10^{-4} h$$

	$Res_{\text{diffraction}}$	Res_{pixel}
0.5 m	$6.1 \cdot 10^{-4} \text{ m}$	$3.140 \cdot 10^{-4} \text{ m}$
1 m	$1.22 \cdot 10^{-3} \text{ m}$	$6.279 \cdot 10^{-4} \text{ m}$
2 m	$2.44 \cdot 10^{-3} \text{ m}$	$1.256 \cdot 10^{-3} \text{ m}$
5 m	$6.1 \cdot 10^{-3} \text{ m}$	$3.140 \cdot 10^{-3} \text{ m}$
200 km	$2.4 \cdot 10^2 \text{ m}$	$1.256 \cdot 10^2 \text{ m}$

Since $Res_{\text{diffraction}} > Res_{\text{pixel}}$ at all distances, this means the resolution is higher at all distances which is worse. Therefore the camera is diffraction-limited.